RAPID STRUCTURED EVIDENCE REVIEW

Published Weather–Health Models

Systematic extraction of selected primary studies, equations and validation evidence

Review status. This is a rapid, purposively scoped evidence review with consistent extraction fields. It is **not an exhaustive systematic review**, a registered protocol, a meta-analysis or a clinical guideline. No original health data or experiments are included. No mathematical model is newly proposed.

Abstract. The objective was to locate published models relevant to thermal tolerance and weather-related cardiovascular, respiratory and symptom outcomes, then separate mathematical specification from empirical validation and personal forecasting. Targeted web searches and citation follow-up were used to verify original papers, public manuscripts, publisher records and author software on 13 October 2026. Twenty-two research, methodological or software publications were selected; one international standard and two reporting guidelines supply context. Extraction recorded outcome, exposure, design, lag, access level, equation location and principal limits. Findings support distinct categories: mechanistic thermoregulation, strain and comfort indices, population count models, distributed lag nonlinear models, case-crossover symptom analyses and thermal-warning applications. An explicit equation or large cohort does not alone establish individual forecast calibration. Reporting inconsistencies were identified in one COPD study. The review provides an auditable reading set and exact equation transcriptions, without claiming complete coverage of all diseases or all publications.

1 Question, scope and eligibility

The question is: *Which previously published mathematical approaches connect meteorological conditions with human thermal response or specified health outcomes, and what evidence supports their use?* The target reader is undertaking literature research rather than fitting a new model.

Included sources had to describe an identifiable human thermoregulation, environmental exposure–response, symptom-analysis or weather-warning method in a published research context. Priority was given to primary model development, primary empirical studies, direct validation and inspectable implementations. Exact-title and DOI verification were used where possible. Sources available only at abstract or metadata level were retained for broad study characteristics but not used to invent missing equations.

Patents, commercial claims without a research article, unsupported universal meteoropathy scores, generated equations and unverified fitted coefficients were excluded from the model evidence. ISO 7933 is retained only as an explicitly identified standard, not counted as a journal validation study [6]. This selection is not a census of cardiology, pulmonology, nephrology or environmental epidemiology.

2 Search and extraction methods

2.1 What was actually searched

Searching consisted of targeted web queries, exact-paper lookups and follow-up through publisher pages, PubMed/PMC-indexed records, university repositories, author-hosted manuscripts and official code pages. These are web-mediated discovery routes, **not documented complete exports from PubMed, Embase, Scopus or Web of Science**. Several PMC pages could not be fully opened in the tool environment; where possible, a publisher or author copy was inspected instead. A source register and representative executed queries appear below.

The search was not prospectively registered and no language-complete or all-year search was performed. No independent human duplicate screening took place. Discovery and extraction were AI-assisted and checked against the cited materials. The corpus was expanded purposively from foundational models to disease-specific examples and a negative validation study. No artificial PRISMA flow diagram, total database hit count or deduplication count is supplied because a reproducible census was not collected. These limitations are disclosed in the spirit of transparent review reporting, not as a claim of full PRISMA compliance [25].

2.2 Extraction and comparison rules

For each selected study, extraction recorded its endpoint, input exposures, population or experimental context, model family, lag where verified, evidence of testing, source access and reproducibility limits. “Not verified” means the reviewed material did not establish the item; it does not prove the original researchers never performed it.

Equation extraction was restricted to readable original mathematical displays or an identified associated implementation. Source numbers, pages or section names accompany transcriptions. Typographical changes such as line breaks do not change the model. Original shorthand and reporting problems are retained and discussed. No missing intercept, pollutant coefficient, error term or individual susceptibility term is inserted by this review.

Appraisal is descriptive rather than a formal GRADE, ROBINS-I or PROBAST assessment. The central distinction is between physiological measurement agreement, observational association, software availability and external prediction evaluation. They are not merged into a numerical “quality score.” No meta-analysis was attempted because outcomes, exposures, reference contrasts and designs are heterogeneous.

3 Included corpus and access audit

The selected research corpus contains **22 publications**: eight papers on thermal models, indices, validation or software, two DLNM methods or software papers, ten disease or symptom applications, and two weather-warning system papers. References [6, 24, 25] provide standards or reporting context and are not added to the primary-study count. This is a count of included publications, **not a search-flow count**.

Access codes. F = relevant full article or author manuscript inspected; A = abstract or bibliographic/institutional record used, without claiming full-method verification; C = associated original code or formula inspected; P = relevant portions of an author-deposited article accessible, but not a complete equation-system audit. A public-source link in the bibliography may offer more access than could be retrieved during preparation.

Ref.	Source / role	Access	Use here
[1, 2]	Fiala / Bröde: thermoregulation and UTCI	A	Scope, not full equations
[3]	Psikuta: Fiala validation	F	Physiological testing
[4]	Takahashi: JOS-3	A+C	Scope and open code
[5]	Malchaire: PHS development / validation	A+P	Algorithm scope and tests
[7]	Moran: physiological strain index	F	Original equation
[8]	Davey: independent PSI limitations	F	Negative validation
[9]	Tartarini / Schiavon: comfort software	A+C	Existing PPD formula
[10]	Gasparrini 2010: DLNM method	F+C	Equations and code
[11]	Gasparrini 2011: R implementation	A	Implementation reference
[12]	Gasparrini 2015: temperature–mortality	A	Design and lag summary
[13]	Bhaskaran: myocardial infarction	A	Registry / method summary
[14, 15]	Alahmad: cardiovascular and stroke mortality	F	Methods and endpoints
[16]	Anderson: respiratory hospitalization	A	Population and model family
[17]	Zhang: adult asthma hospitalization	F	Equation and adjustments
[18]	Li: COPD hospitalization	F	Equation; reporting caution
[19]	Mukamal: severe headache	A	Design and endpoint
[20]	Dixon: daily pain	F	Within-person design
[21]	Bobb: cause-specific admissions	F	Matched design / disagreement
[22, 23]	Kingma / Morabito: warning systems	A	Architecture scope only

4 Thermal-model extraction

4.1 Fiala, UTCI and validation

Model descriptions [1, 2]; independent physiological evaluation [3].
Endpoint. Fiala predicts physiological thermal responses; UTCI expresses an equivalent thermal environment. Neither output is an individual disease probability.
Inputs and structure. The thermal setting incorporates temperature, humidity, air movement and radiation, with modeled physiological regulation. The full coupled equations were not re-extracted from the unavailable accepted-manuscript download.
Evidence. The accessible validation article compares modeled and experimental physiological measurements. This supports assessment of body-temperature predictions under tested conditions, not direct clinical event forecasting.
Reproduction boundary. Obtain the complete published system or implementation. A generic whole-body balance would not be a faithful replacement for the model.

4.2 JOS-3

Published model and associated public implementation [4].
Endpoint. Segmented body temperatures and thermoregulatory responses.
Specification. An 83-node model with backward-difference numerical computation is described in the publication record; the authors provide code with personal and environmental input settings.
Use and limit. Suitable as an inspectable mechanistic source. The code's existence does not demonstrate an externally validated heart-attack, COPD or asthma predictor. Code version and parameter choices remain part of any later reproduction.

4.3 Predicted heat strain

Malchaire et al. [5]; standard context [6].
Endpoint. Sweat loss, rectal temperature and exposure-duration assessment based on specified strain criteria.
Evidence. The development/validation publication reports 909 laboratory and field experiments.
Important distinction. The 2001 paper and ISO 7933:2023 are related but different sources. This review does not claim to reproduce the entire current standard or compute a patient-specific allowable duration.

4.4 PSI, its later validation limits, and PPD

Original PSI [7]; later experimental critique [8]; comfort implementation [9].
Endpoints. PSI combines measured rectal temperature and heart rate. PPD estimates thermal dissatisfaction from PMV. Their units and intended meanings differ.
Equation availability. Both compact relations are reproduced in Section 7. PSI's normalization constants are not personal safety guarantees; PPD is not an illness probability.
Contrary evidence. Davey et al. found that absolute PSI did not reliably identify thermal tolerance limits under their experimental conditions. This source is retained rather than presenting only supportive development results.

5 Disease and symptom study extraction

5.1 Myocardial infarction

Bhaskaran et al., BMJ 2010 [13].
Data and outcome. 84,010 myocardial infarction admissions; 15 conurbations in England and Wales.
Exposure and model. Temperature-related time-series regression with effects evaluated over the current and following 28 days.
Interpretation. The reported association was primarily with lower temperature. The study is evidence about a defined clinical event and setting, not a general equation for all cardiovascular illnesses. Full fitted coefficients were not reconstructed from the abstract.

5.2 Cardiovascular cause-specific mortality

Alahmad et al., Circulation 2023 [14].
Data and outcome. 567 cities in 27 countries; total cardiovascular and selected cause-specific deaths.
Model and lag. City-specific count models, temperature/lag cross-bases and hierarchical meta-regression. The principal exposure–response basis uses quadratic B-splines; lags extend to 14 days, with a natural spline for the lag response.
Stratification. Cause of death and geographic structure are represented. Pooling across cities and climate/country groups is not person-specific learning.
Validation boundary. An extensive multisite association analysis, not an external prospective test of individualized next-day event probabilities.

5.3 Ischemic and hemorrhagic stroke mortality

Alahmad et al., Stroke 2024 [15].
Outcome. Stroke deaths separated into ischemic and hemorrhagic subtypes.
Model. Conditional quasi-Poisson regression, time-stratified control through year, month and day-of-week structure, nonlinear distributed lags and pooling. Cumulative effects were assessed up to 14 days.
Limit. Mortality is not stroke incidence. The model does not supply a universal absolute-risk equation for a person with previous stroke.

5.4 All-cause temperature–mortality analysis

Gasparrini et al., The Lancet 2015 [12].
Data and exposure. 384 locations; ambient temperature and observed daily mortality.
Model. Location-specific time-series Poisson models and DLNM with a 21-day lag, followed by multivariate meta-regression.
Use. An empirical application of nonlinear and delayed effects across climates. Its reference temperatures, coefficients and 21-day window should not be transferred without justification to a different clinical outcome.

5.5 Respiratory hospitalizations in older adults

Anderson et al., 2013 [16].
Data. Approximately 12.5 million Medicare beneficiaries, 213 US counties, 1999–2008.
Model and outcome. Bayesian hierarchical estimation of heat-related emergency respiratory hospitalization risk, including same-day associations.
Limit. The population and diagnostic grouping are specific. No exact regression expression is transcribed because only the publication's abstract-level methods were verified here. This is not a personally calibrated asthma forecast.

5.6 Adult asthma admissions

Zhang et al., PLOS ONE 2014 [17].
Data. 15,678 adult admissions in Shanghai, 2005–2012; daily medical-insurance admission counts and fixed-site environmental observations.
Specification. Semiparametric count-model analysis with a log link and a temperature cross-basis, with maximum lag 30 days. Humidity, rainfall, wind, season/time, holidays, day of week and pollution are represented.
Equation. The original unnumbered display on PDF p. 2 is reproduced in Section 7. Its abbreviated pollutant term is preserved.
Evidence boundary. Residual checks and sensitivity to model settings assess aspects of the fitted analysis, not external operational forecasting performance.

5.7 COPD admissions

Li et al., Scientific Reports 2025 [18].
Data. 12,841 admissions, Qingyang, China, 2015–2019.
Specification. Separate quasi-Poisson GAM/DLNM analyses for temperature, diurnal temperature range and relative humidity. Reported maximum lags are 21 days for temperature/range and 7 days for humidity. Age and sex groups are examined.
Reporting concerns. The formulas use $\log(Y_t)$ despite a count-mean modeling description. Pollutants appear explicitly in the model expressions, but the limitations say air-pollution confounding was not adjusted for. These are unresolved internal inconsistencies, not errors silently corrected by the review.
Use. An inspectable published specification, but not a preferred unquestioned template for exact reproduction. No universal COPD coefficient set follows from it.

5.8 Severe headache presentations

Mukamal et al., Neurology 2009 [19].
Data and endpoint. 7,054 emergency presentations for severe headache; Boston study.
Design. Case-crossover comparisons of short-term weather and pollution exposures, including temperature, pressure and humidity.
Access and limit. The indexed methods/abstract and correction record were verified; an equation is not reconstructed from them. Emergency presentation is not equivalent to every migraine episode or every symptom reported at home.

5.9 Daily pain symptoms

Dixon et al., npj Digital Medicine 2019 [20].
Data and endpoint. Repeated reports from 2,658 participants over 15 months. The study identifies clinically relevant increases in self-reported pain.
Design and inputs. Case-crossover analysis linked to weather using participant location. Temperature, humidity, pressure and wind were considered, with additional examination of mood and activity.
Interpretation. Modest associations and repeated personal data do not imply a separately trained, calibrated prediction model for every participant. The original study is not described here as a newly invented mixed-effects model.

5.10 Renal and other heat-related admissions

Bobb et al., JAMA 2014 [21].
Design. Matched comparison of heat-wave and non–heat-wave periods in 1,943 US counties. Heat-wave exposure involved at least two consecutive days above a county-specific 99th-percentile threshold.
Outcomes. Higher risks were identified for fluid/electrolyte disorders, renal failure, urinary tract infection, septicemia and heat stroke. Congestive-heart-failure admissions were lower; the analysis did not reproduce the earlier respiratory result under its definitions.
Interpretation. This is a reason to preserve outcome and exposure definitions, not evidence that one contradictory result must be discarded. Hospitalization and death are different endpoints.

6 Warning-system studies and scope of personalization

Kingma et al. describe ClimApp, and Morabito et al. describe HEAT-SHIELD [22, 23]. Both belong to the selected published system literature because they connect forecasts with personal or occupational information. The access used here supports an architectural comparison, not a full audit of every application algorithm.

Their relevance is practical: personalization of thermal exposure and warnings has precedents. Their inclusion does not establish validated forecasts for all pre-existing diseases. This review reports no new combined warning equation and does not substitute a patent or product description for disease-specific clinical evaluation.

7 Source-verified equation register

The following displays are transcriptions rather than new models. The full physiological differential-equation systems of Fiala, JOS-3 and PHS are not shortened into substitute formulas. Numeric constants, original shorthand and source locations are retained.

7.1 Physiological strain index

$$\mathrm{PSI} = 5\frac{T_{\mathrm{re},t} - T_{\mathrm{re},0}}{39.5 - T_{\mathrm{re},0}} + 5\frac{\mathrm{HR}_t - \mathrm{HR}_0}{180 - \mathrm{HR}_0}. \tag{1}$$

Source: Moran et al. [7], original Eq. (4); also explicitly reproduced under “Calculations / Physiological strain index” in Davey et al. [8]. Fractions replace algebraically identical inverse factors; the model is unchanged.

T_{re} is rectal temperature in degrees Celsius; HR is heart rate in beats per minute. Subscript 0 denotes baseline and t the measurement time. Interpretation must include the later negative evaluation [8].

7.2 Thermal dissatisfaction

$$\mathrm{PPD} = 100 - 95 \exp\left(-0.03353\,\mathrm{PMV}^4 - 0.2179\,\mathrm{PMV}^2\right). \tag{2}$$

Source: The public `pmv_ppd_iso` implementation associated with *pythermalcomfort* [9], specifically the assignment to `ppd_array`. This is an existing implemented relation, not a new fit or a numbered equation attributed to the software article.

This relation maps PMV to modeled dissatisfaction; it does not map weather to clinical disease probability.

7.3 General mean model and DLNM cross-basis

$$g(\mu_t) = \alpha + \sum_{j=1}^{J} s_j(x_{tj}; \boldsymbol{\beta}_j) + \sum_{k=1}^{K} \gamma_k u_{tk}. \tag{3}$$

Source: Gasparrini, Armstrong and Kenward [10], original Eq. (1), p. 2225.

$$s(x_t; \boldsymbol{\eta}) = \sum_{j=1}^{v_x} \sum_{k=1}^{v_\ell} \mathbf{r}_{tj\cdot}^{\mathsf{T}} \mathbf{c}_{\cdot k}\, \eta_{jk} = \mathbf{w}_{t\cdot}^{\mathsf{T}} \boldsymbol{\eta}. \tag{4}$$

Source: Gasparrini, Armstrong and Kenward [10], original Eq. (7), p. 2227. Boldface and transpose symbols are typographical renderings of the original vector notation.

In the original notation, $\mu_t = E(Y_t)$, g is the link, and s_j is an exposure-response component. The cross-basis combines exposure and lag transformations. $\mathbf{r}_{tj\cdot}$ and $\mathbf{c}_{\cdot k}$ refer to those basis components; η is estimated. The software source in [11] supports implementation, not an additional independent clinical validation.

7.4 Adult asthma study

$$\begin{aligned} \mathrm{Log}[E(Y_t)] = \alpha &+ \gamma\,\mathrm{DMT}_{t,l} + s(\mathrm{time}, \mathrm{df}) + s(\mathrm{humidity}, \mathrm{df}) \\ &+ s(\mathrm{rainfall}, \mathrm{df}) + s(\mathrm{wind\ speed}, \mathrm{df}) \\ &+ \beta\,\mathrm{Holiday} + \varepsilon\,\mathrm{DOW} + Z_t. \end{aligned} \tag{5}$$

Source: Zhang et al. [17], "Data analysis," unnumbered basic-model equation, PDF p. 2. Line breaks only are editorial. The original pollutant shorthand Z_t is retained.

Y_t is the daily count; DMT is the temperature cross-basis; s denotes spline adjustment. Z_t is the source's abbreviated pollutant contribution. The printed εDOW is retained as a calendar term rather than reinterpreted as random noise.

7.5 COPD temperature model: original notation retained

$$\begin{aligned} \log(Y_t) = \alpha &+ \beta\,\mathrm{TEM}_{t,l} + \mathrm{ns}(\mathrm{PM}_{10t}, \mathrm{df}) \\ &+ \mathrm{ns}(\mathrm{SO}_{2t}, \mathrm{df}) + \mathrm{ns}(\mathrm{NO}_{2t}, \mathrm{df}) \\ &+ \mathrm{ns}(\mathrm{RH}_t, \mathrm{df}) + \mathrm{ns}(p_t, \mathrm{df}) \\ &+ \mathrm{ns}(\mathrm{Time}_t, \mathrm{df}) + \mathrm{Dow} + \mathrm{Holiday}. \end{aligned} \tag{6}$$

Source: Li et al. [18], "Statistical models," first unnumbered formula. The printed left side $\log(Y_t)$ is preserved, not silently changed to $\log E(Y_t)$. See the reporting cautions accompanying this transcription.

ns denotes natural cubic splines, TEM the temperature cross-basis, RH relative humidity and p_t pressure. The source's two additional models replace the primary exposure with diurnal temperature range or humidity and alter the adjustments. No one-line replacement or corrected model is invented here. The count-mean notation and pollutant-reporting inconsistencies described above remain unresolved.

8 Cross-study appraisal

8.1 Specification is not validation

The clearest compact equations in this set are not necessarily the strongest personal safety tools. PSI is explicit yet has a later adverse validation finding. Conversely, a large multicountry disease study can have persuasive association evidence while lacking a deployable individual probability model. Code availability, numerical transparency, physiological agreement and clinical forecast calibration should therefore be recorded separately [8, 10, 14, 24].

8.2 Population estimates do not establish individual absolute risk

The clinical studies estimate different objects: admission associations, mortality relationships or symptom-event odds. None of the extracted evidence establishes one fixed equation that turns tomorrow's weather and arbitrary disease history into accurately calibrated probabilities for every condition. This is a conclusion about this selected corpus, not a proof that individual prediction is impossible.

Where age, sex or geographic groups are studied, subgroup analysis should not be relabeled as unique-person learning. Where physiological constants occur, they should not be repurposed as universal clinical cutoffs. These interpretation rules avoid converting a genuine published model into a different, unvalidated one.

8.3 No common effect-size or time-limit scale

A core-temperature error, an index value, an admission relative risk and an odds ratio cannot be averaged into one "weather danger" metric without developing a new model. Likewise, lags of 7, 14, 21, 28 or 30 days represent study-specific temporal relationships, not interchangeable permissible exposure durations. This review intentionally provides no new integrated score.

8.4 Credibility problems and disagreements are retained

The COPD article's internal reporting inconsistency is disclosed beside its equation [18]. The later PSI findings are presented alongside the development paper [7, 8]. The different respiratory and heart-failure findings across admission and mortality studies are not hidden [16, 21, 14]. These comparisons are qualitative appraisals, not a formally validated risk-of-bias instrument.

9 Limitations and implications for further reading

This review is limited by targeted rather than exhaustive searching, heterogeneous article access, no independent screening, no protocol registration, and no formal risk-of-bias assessment. Search results depend on indexing and retrieval, and articles beyond the selected corpus may change an endpoint-specific conclusion. Full supplements, source code and data were not executed as a reproduction study. Sources dated after the foundational papers were included when relevant, but completeness through 2026 is not claimed.

The literature set supports several precise next reading tasks: use PHS and its standard for exposure-duration methodology; read Fiala alongside its validation and JOS-3 alongside its code; use Gasparrini's original algebra and software for delayed exposure modeling; and retain the clinical endpoint and population when reading individual disease applications. These are routes through existing evidence, not new mathematical proposals.

A full systematic review would require a prospectively specified question, reproducible database searches and eligibility decisions, a complete screening trail, appropriate appraisal, and a planned synthesis. PRISMA is a reporting framework rather than a guarantee that these steps have been done [25]. This document supplies systematic extraction of selected sources but does not claim that broader status.

10 Conclusions

Selected peer-reviewed literature establishes several distinct mathematical approaches to weather and human health. Their scope extends from physiological heat response to cause-specific mortality, admissions and pain. Exact source formulas can be transcribed and compared without original experiments, but their fitted coefficients, intended population, assumptions and validation outcomes must remain attached. The present evidence review introduces no new equation and establishes no universal clinical weather-risk or personal endurance formula.

A Representative executed search queries

These strings document a subset of the actual discovery queries used on 13 October 2026. They are not presented as an exhaustive database strategy or a recoverable total-hit ledger. Exact-title opening, DOI checking and citation follow-up supplemented these searches.

ID	Executed web query	Purpose
Q1	“The physiological strain index does not reliably identify individuals” Davey 2021	Contrary validation
Q2	“Heat-related emergency hospitalizations for respiratory diseases” 2013 Anderson	Respiratory study
Q3	“Cause-specific risk of hospital admission related to extreme heat” Bobb 2014	Other diagnoses
Q4	“Extreme Temperatures and Stroke Mortality” 2024 Alahmad	Stroke subtype study
Q5	“ClimApp” “2021” “Kingma”	Published warning system
Q6	“An Occupational Heat–Health Warning System for Europe” Morabito 2019	Occupational warning
Q7	“Davey” “physiological strain index” “2021” “10.1007”	Metadata cross-check
Q8	“Thermoregulation model JOS-3” “Yoda”	Model authors / code
Q9	“Validation of the Fiala” “443”	Validation citation
Q10	“Mortality risk attributable” “2015” “384”	Multicountry method

Equation-check routes. The original Gasparrini PDF was inspected at pp. 2225 and 2227; the Zhang PDF at p. 2; Li’s publisher HTML under “Statistical models”; Moran’s original equation and Davey’s published reproduction; and the `ppd_array` expression in the official comfort-model implementation. Links are attached to the corresponding bibliography entries rather than inserted into the body.

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